

# LEAD EVIDENCE PACK — KIMI independent review of LPW-CAND-20260912

Reviewer: lead coordinator (KIMI line). Exposure: the full candidate and review packet were read before review — this is ordinary hostile review, not blind reconstruction. All computations below are independent of the author's `verify_local_path.py` lineage (sympy/mpmath, separate process).

## A. Algebraic core — INDEPENDENTLY VERIFIED (sympy, exact rational arithmetic)

- $F(x,y) = x^3/3 - x/4 - 1/12 + 2(x^2-1/4)y - 5y^2$ :  $F(m) = 0$ ,  $F(s) = -1/6$ ,  $F = 0$  at both  $(m = (-1/2, 0), s = (1/2, 0))$ ;  $H(m) = [[-1, -2], [-2, -10]]$ ,  $\det 6$  (negative definite);  $H(s) = [[1, 2], [2, -10]]$ ,  $\det -14$  (saddle). VERIFIED.
- Path height  $R(x) = F'(x) = (2x+1)^2(12x^2+8x-17)/240$  identically;  $R(x) + 99/1280 = (4x+5)^2(48x^2-40x+1)/3840$  identically;  $48x^2-40x+1 > 0$  on  $[-2, -1/2]$  (roots 0.026..., 0.807...);  $\min R = -99/1280$  at  $x = -5/4$ ;  $R(-2) = 9/16$ . VERIFIED.
- Clearance:  $1/6 - 99/1280 - 1/16 = 640/3840 - 297/3840 - 240/3840 = 103/3840 > 0$ . VERIFIED (an early partial check of mine misread the display; the candidate's arithmetic is correct).
- $C^2$  perturbation ( $\epsilon = 1/16$ ):  $\det H_G(m) = (1-\epsilon)(10-\epsilon) - (2+\epsilon)^2 = 81/16 > 5$ ;  $\det H_G(s) = -(1-\epsilon)(10-\epsilon) - (2-\epsilon)^2 = -1673/128 < -13$ ;  $|\det \text{product}| = 65$  (actual 66.17); endpoint  $G = 9/16 - 1/16 = 1/2 > 0$ ; path  $\min = -179/1280 > -1/6$ . VERIFIED.

## B. Hermite normalization — INDEPENDENTLY VERIFIED (sympy)

- $V$  (row-major evaluations at  $m, s$  on  $(1, x, y, x^2, xy, x^3)$ ):  $\det V = -1$ . VERIFIED.
- Displayed  $T_r$ :  $\det = -r^5$ , invertible for all  $r > 0$ . VERIFIED.
- $u_r = T_r(b, 0, 0, b-r^3/6, 0, 0) = (b - r^3/12, -r^2/4, 0, 0, 0, 1/3)$ . VERIFIED.
- Each row of  $T_r$  reproduces the claimed derivative-frame limit with trapezoidal corrections:  $(f(M) + f(S))/2 + (r/8)(f_x(M) - f_x(S)) \rightarrow f(0) + O(r)$  (the  $r^2$  terms cancel exactly);  $(3/(2r))(f(S) - f(M)) - (1/4)(f_x(M) + f_x(S)) \rightarrow f_x(0) + O(r)$  (exact through cubics);  $(f_x(S) - f_x(M))/(2r) \rightarrow f_{xx}(0)/2 + O(r^2)$ ;  $(f_y(S) - f_y(M))/r \rightarrow f_{xy}(0) + O(r^2)$ ;  $(2/r^3)(f(M) - f(S)) + (1/r^2)(f_x(M) + f_x(S)) \rightarrow f_{xxx}(0)/6$  exactly ( $-f_{xxx}/12 + f_{xxx}/4 = f_{xxx}/6$ ). VERIFIED.
- Note: the formula " $T_r = D_r \{^{-1}\}^V R_r$ " holds with the ROW-major  $V$  (the reconstruction  $R_r \cdot T_r \{^{-1}\}^{D_r}$  is exactly the row-major evaluation matrix,  $\det -1$ ); a column-major reading fails — the displayed matrix is the correct object, and the formula is consistent under the row-major reading. Noted for the record (cosmetic; no mathematical content affected).

## C. Taylor lift coefficients — INDEPENDENTLY VERIFIED (hand derivation, term-by-term)

- $|f_{xx}(0)| = Kr^2/24$  (subtract endpoint gradient equations; remainders  $Kr^3/48$  each);  $|f_x(0)/r^2 + f_{xxx}(0)/8| = Kr/48$  (add them);  $|f_{xxx}(0) - 2| = Kr/4 + Kr/16 = 5Kr/16$  (value difference + substitution);  $|(f(0)-b)/r^3 + 1/12| = Kr/192 + Kr/384 = Kr/128$ ;  $|f_x(0)/r^2 + 1/4| = 5Kr/128 + Kr/48 = 23Kr/384$ ;  $|f_y(0)/r^2 + f_{xy}(0)/8| = Kr/48$  (the  $y$ -gradient equation — the term completing the sum); remainder  $(9/2)Kr$ ; monomial  $C^2$  bounds on  $D$ :  $1 \rightarrow 1$ ,  $x \rightarrow 2$ ,  $y \rightarrow 1$ ,  $x^2 \rightarrow 4$ ,  $xy \rightarrow 2$ ,  $x^3 \rightarrow 12$ . Total:  $3 + 46 + 8 + 32 + 32 + 240 + 1728 = 2089/384$  exactly,  $< 8K$ . VERIFIED.
- Tolerance:  $16 + 8Kr = 16/1024 + 8/256 = 3/64 < 1/16$  at  $r = \min(r_G, 1, (256K)^{-1})$ . VERIFIED.

- Hessian scaling  $H_{f(rz)} = r \cdot H_{\{F_r\}}(z)$ ; det scales by  $r^2$  per station; pair by  $r$ ; on the event  $W_r$   $(81/16)(1673/128) r^{-65r}$ . VERIFIED.
- Physical path:  $\min f = b - (179/1280)r^3 > b - r^3/6 = f(S)$  ( $179 \cdot 6 = 1074 < 1280$ ); endpoint  $f|_b + r^3/2 > b$ ;  $rD$  — one torus chart for  $r \leq 1$ . VERIFIED.

## D. Ten-jet Gaussian limit — NUMERICAL EVIDENCE (exact field, mpmath dps 60, |k| = 30)

- Covariance of the ten-jet  $(U_0, J)$  at the origin: positive definite, min eigenvalue 0.12655 (computed from exact lattice sums; truncation tail  $< 1e-60$  at  $KMAX = 30$ ).
- Schur complement  $\text{Cov}(J | U_0) = \text{diag}(2, 1/2, 1/2, 1/6)$  at machine precision (exact rationals).
- Conditional mean  $E[J | U_0 = u_0] = (-b, 0, 0, 0)$  at  $u_0 = (b, 0, 0, 0, 0, 1/3)$ .
- Gaussian density of  $J$  at the thin-box center ( $r = 0.025$ ,  $q = -10r$ ,  $A = 2$ ,  $B = D_3 = 0$ ):  $1.28e-3$ , and  $1.29e-3$  at the box corners — concrete positivity of the density the candidate bounds below by  $m$  (the uniform-in- $r$  argument is continuity + compactness; this display is evidence at the  $r \rightarrow 0$  endpoint, not a proof of uniformity).
- Structural note (supports the power ledger): on the thin event both diagonal Hessian entries are  $O(r)$  ( $f_{yy} = -10r$ ), so  $|\det H|$  per station is  $O(r^2)$  and the weight  $65r$  is  $r^2$  times the generic typed  $O(r) \cdot O(r)$  — consistent with  $Z_r = C_Z r^2$  and the quotient  $r^{\{1+4-2\}} = r^3$ .

## E. Estimand cross-check (against the program's records)

- The candidate's pins ( $f(M) = b$ ,  $f(S) = b - r^3/6$ ,  $f = 0$  at  $M = (-r/2, 0)$ ,  $S = (r/2, 0)$ ) are exactly the program's typed pair-Palm pins (manuscript Definition 2.8; the C024/C025 construction: " $M = (-r/2, 0)$ : jet  $(b, 0, 0)$ ;  $S = (r/2, 0)$ : jet  $(b - r^3/6, 0, 0)$ ",  $= r^3/6$ ).
- The candidate's  $Q_r$  spans the same conditioning as the manuscript's corrected pin basis  $V_r$  (Preliminary 2.5) and as the raw six pins (all three bases span the raw six-pin space); the  $T_r$  transformation is a third basis of the same span. Same Gaussian conditioning.
- Weight  $W_r = |\det H_M \det H_S| \cdot 1_{\{H_M \leq 0, \det H_S < 0\}}$  matches the manuscript's  $W_r = |\det H_M \det H_S| \cdot \text{indicators}(M \text{ max}, S \text{ saddle})$ ;  $Z_r = r^2$  consistent with the campaign's G.7 normalizer theorem ( $c_Z r^2 = Z_r = C_Z r^2$ ); the candidate needs only the upper bound and positivity (the constructed event supplies  $Z_r > 0$ ).
- $q(r, b) = P_r\{D_f(M) = S\}$  matches W10's estimand  $q(r, 6/5) = p_r = P_r\{D(M) = S\}$  (no adjacency conditioning; the  $_r$  channel is untouched — the candidate's event implies failure of the typed elder pairing directly).
- NOT locally verifiable: the byte-level match with GP-DER-118-v1.10 Section 0 (a Drive-side file not in the local corpus; the author's package reports its hashes reverified: whole-file c1d6e559..., body 9b7901e1...). Scoped as CANNOT-VERIFY-locally; the conventions are consistent with every record in the local corpus.

## F. Topology — the lead's own attempt to break it

Attempted obstruction: an elder-rule interpretation where  $M$  still dies at  $S$  despite the path. Analysis: at level  $h = (s, \min f(\cdot))$ ,  $M$  and  $z$  ( $f(z) > b$ ) share a superlevel component; that component's birth is  $f(z) > b = \text{birth}(M)$ , hence strictly older. At level  $s$ , the saddle  $S$  merges components;  $M$ 's class is not the oldest present, so  $M$  does not die at  $S$  (it may have died earlier — also a failure of  $D(M) = S$ ). The argument is pathwise: no Morse–Smale, no gradient trajectories, no count of any third saddle. The empty-birth convention ( $\sup = -\infty$ ) does not touch it. The one requirement is the elder rule itself (already part of

the estimand's definition). UNBROKEN.

## G. Remaining scrutiny (for the record)

- The conditional  $C$  moment (R3) is the candidate's most delicate step; RC's independent referee pass is in flight. The lead's own read: the regression-representer argument is sound at the proof level (representers'  $C$  norms uniform by Cauchy–Schwarz + covariance convergence; residual's moments  $v$ -independent and finite by Minkowski;  $\Gamma_{\hat{r}^{-1}}$  bounded on  $[0, r_G]$ ; the regression version makes the pointwise-in- $v$  Markov inequality exact rather than a.e.), matching the standard Palm-Kac–Rice technique for Gaussian fields.